

NAPOT Open-Data Component Gate

What an Adverse Receiver-Relative Phase Result Permits, Blocks, and Requires Next

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Abstract

A visual system receives incomplete signals from many locations and must continually turn them into useful predictions. A proposed timing-sensitive mechanism should therefore earn its place by improving prediction beyond an equally capable model that receives the underlying signals without the proposed transformation. This paper reports the decision consequences of one such test. It is a governed program update built from a previously executed development analysis, not a second experiment or an independent replication.

The analysis used the Allen Institute Visual Coding Neuropixels dataset. It tested whether a 25-feature receiver-relative representation of phase, amplitude, event duration, and spatially normalized lag improved prediction of the next 20-ms VISp spike count for an anatomically declared LGd-to-VISp edge. Fourteen coordinate-eligible sessions from 14 unique specimens contributed 921 units and 908,295 eligible 20-ms time bins. Each branch received the same 16 random features derived from 14 raw lag variables and had the same output capacity. The raw branch added eight independently projected raw features; the proposed branch added eight projections of the 25-feature receiver-relative encoding. The capacity-matched raw branch performed better in every analyzed session. The cross-session median raw-minus-proposed mean Poisson-deviance difference per held-out time bin was -0.001099, with a descriptive session-bootstrap 95% interval from -0.001301 to -0.000872.

Matched perturbations locate the failure more precisely without rescuing the losing branch. The intact feature family performed better than circularly shifted and source-receiver-swapped versions, and better than a version with amplitude-duration terms removed. Those are within-model sensitivities, not proof of anatomical direction, causal transfer, or incremental value beyond raw signals. Removing the configured phase terms improved prediction in all 14 sessions. The tested implementation therefore fails its development gate. The executed code also omitted the amplitude, valid-cycle, and circular-concentration phase-quality mask described in the parent protocol. The result applies to the implementation that actually ran; it neither tests nor rescues a quality-gated successor. This does not contradict preparation-specific evidence that phase-of-firing can add visual information, and it does not test the full recurrent Neural Array Projection Oscillation Tomography architecture, behavior, the returned sensory consequences of action (reafference), perceptual content, or consciousness. Five final-test sessions remain sealed. The warranted next step is to preserve this adverse result, separate the potentially useful non-phase components, and prospectively freeze any redesigned estimator before opening another endpoint.

Keywords: Neural Array Projection Oscillation Tomography; NAPOT; phase-wave differential; open data; Neuropixels; visual cortex; negative result; model comparison; research program governance

Scope and evidence boundary

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1. The problem before the vocabulary

1.1 An ordinary prediction problem

No visual neuron receives a complete scene. Different populations receive different features, positions, delays, and mixtures of sensory and internally generated activity. Yet the system must still predict what is likely to happen next. A candidate mechanism matters scientifically only if its special transformation contributes something that a fair alternative cannot recover from the underlying signals.

Suppose two populations have similar mean firing rates and spectral power. Their downstream consequences may still differ because their events arrive at different points in a receiver's current cycle, last for different intervals, or participate in different spatial patterns. That possibility motivates a timing-sensitive representation. It does not establish that the representation works. The operational question is:

After capacity, training data, raw inputs, target, and evaluation procedure are matched, does the added receiver-relative representation improve held-out prediction?

This question must be answered before the representation is used as evidence for a larger theory.

1.2 What established neuroscience already supplies

Phase-sensitive coding is not an empty research direction. Montemurro and colleagues reported that, in anaesthetized macaque primary visual cortex viewing a naturalistic movie, low-frequency phase-of-firing carried information beyond spike count [1]. That result establishes that phase can add visual information in a particular preparation and coding analysis. It does not guarantee that every phase estimator, species, pathway, target, or model comparison will succeed.

Phase analyses also have known failure modes. Nonsinusoidal waveform shape and nonstationarity can create misleading phase or cross-frequency structure [2,3]. Connectivity estimates can be distorted by source mixing, volume conduction, noise, and finite-sample bias [4,5,7]. Filtering can alter waveform timing and shape [6]. These are reasons to demand explicit estimators, nulls, and matched comparators rather than reasons to dismiss temporal structure in advance.

1.3 The proposed operation in plain language

The tested operation asks how a putative sender-receiver relationship at the present moment differs from its own strictly earlier context. It records four kinds of departure:

1. a wrapped phase departure;
2. a log-amplitude departure;
3. the causal duration of the present event;
4. a lagged phase difference normalized by declared spatial distance.

The receiver matters because the same arriving pattern may have different consequences in populations with different recent histories, inhibition, morphology, or excitability. The past-only baseline matters because the event being evaluated cannot be allowed to define its own expectation. The matched raw comparator matters because a transformed representation has not added value if an equally sized raw branch predicts the target better.

Only after this operation is understood is the name useful. The four-part candidate is called a **phase-wave differential** (PWD). It is one possible component of **Neural Array Projection Oscillation Tomography** (NAPOT), a larger proposal in which distributed populations receive partial conditions, transform them in their current context, project consequences, and recurrently update an action-relevant estimate.

2. The larger architecture and this paper’s ownership boundary

Let population a observe only part of a latent state s_t :

$$y_{a,t} = O_a(s_t) + \epsilon_{a,t}, \quad (1)$$

where O_a is a partial observation operator and $\epsilon_{a,t}$ collects measurement error and unmodeled variation. A candidate recurrent system combines directed population projections with its prior estimate and any earlier action:

$$\hat{s}_t = R_\theta \left(\{p_{a,\leq t}\}_{a=1}^A, u_{<t}, \hat{s}_{t-1} \right). \quad (2)$$

Equation (2) is a research target, not a result of the present component test. It earns support only if the inferred state predicts held-out neural or behavioral variables, survives matched alternatives, and responds to interventions as prospectively predicted.

Here $p_{a,\leq t}$ denotes the history of directed projections supplied by population a through time t , while $u_{<t}$ denotes actions taken strictly before time t . The reconstructed state $\hat{s}_t$ is therefore conditioned on partial population evidence, prior action, and the preceding estimate rather than on an unobserved all-seeing input.

Four companion manuscripts have distinct roles:

- The NAPOT ten-stage methods paper owns the recovered genealogy, synthetic calibration, and the unopened future M0-M7 recurrent benchmark [9].
- The receiver-relative PWD paper owns the software operator and the sole 14-session empirical execution record [10].
- The PWD calculus paper owns the typed calculus, neural-tuning, variability-token, and Taylor-sequence formal program; it does not own this empirical result [11].
- This paper owns the NAPOT-level inference boundary: what the one adverse execution permits, blocks, and requires next.

This paper is therefore a concise program update and component-gate interpretation. No new model was fitted and no new dataset was analyzed for this paper. It is not a second dataset, a second empirical result, or an independent replication of the companion empirical paper. Its distinct contribution is to prevent a narrow failure from being inflated into total falsification while also preventing surviving ablations from being inflated into support for the full theory.

3. Dated Micah genealogy and its evidential boundary

The historical record establishes why this research program asked about changing modes of neural communication. It does not determine the biological result.

Table 1. Dated source ladder for the bounded NAPOT / PWD research question.

Stage	Source and custody	Bounded contribution
2011 owner-dated ancestor, publicly fixed in 2022	a0348z, internally dated May 13, 2011 and first publicly Git-fixed June 8, 2022 [12]	Describes distributed scene fragments or “pixels,” learned statistical links, prediction from partial patterns, and a continuously updated multisensory picture. This is an early partial Neural Rendering / NAPOT ancestor, not the mature PWD vector or a 2011 public-priority date.
2017 public ancestor	<i>The Neural Lace Podcast Origin: The Neo Mind Cycle Concept</i> , uploaded April 11, 2017 [13]	Asks whether neural signaling alternates between connection-and-feedback and connectionless modes and proposes the Alternating Transmission Protocol. This is a protocol-search ancestor, not mature PWD or NAPOT.
2022 immutable public fixation	SAN Git snapshot 16f895fb935933a4d523a29e2ec27a619a91b6ae [12]	Fixes the then-current orientation set and the August-November 2022 NAPOT 2-6 changelog. It establishes public development of the architecture, not empirical validation.
2025 public paper	<i>Self Aware Networks</i> on Zenodo [8]	Publicly fixes the broader SAN architecture and terminology. The exact provider title is <i>Self Aware Networks</i> .
2026 current synthesis	NAPOT, PWD, and PWD-calculus companion manuscripts [9-11]	Formalizes the test program and current distinctions. These private successors cannot be used as earlier public proof.

The 2011 date is author text preserved inside a file first publicly fixed by Git in June 2022; those date types are not interchangeable. The 2017 audio has a verified public upload timestamp. Exact transcript wording is separately fixed no later than the immutable 2022 Git transcript. The mature phase-specific operator must not be backdated into either earlier ancestor. Conversely, the adverse 2026 development result does not erase the earlier authorship record.

4. The tested receiver-relative component

For a declared edge $e = (a \rightarrow b)$, frequency center f , and time t , let p_{eft} , α_{eft} , ℓ_{eft} , and g_{eft} denote the four receiver-relative components. Collecting each component across the five frozen centers in the implementation’s family-major order gives

$$\mathbf{x}_{et}^{\text{PWD}} = \begin{bmatrix} \sin \mathbf{p}_{et} \\ \cos \mathbf{p}_{et} \\ \vec{\alpha}_{et} \\ \log(1 + \vec{\ell}_{et}) \\ \mathbf{g}_{et} \end{bmatrix} \in \mathbb{R}^{25}, \quad (3)$$

where each vector-valued row contains five entries indexed by f : p is wrapped phase departure from a strictly past circular baseline, α is receiver log-amplitude departure from a strictly past reference, ℓ is causal event duration in samples, and g is a coordinate-normalized lagged phase-gradient term [10]. Sine and cosine jointly encode the circular phase variable without a discontinuity at $\pm\pi$. The five filter centers are the geometric centers of the nominal ranges 4-8, 8-15, 15-30, 30-55, and 65-100 Hz; the code did not implement

five full conventional bandpass filters. Stacking five component families across five centers produces 25 inputs. The calculation uses no future sample.

The parent protocol also described a phase-quality indicator requiring frozen amplitude, valid-cycle-count, and circular-concentration criteria. Neither the imported causal operator nor the grouped-session runner implemented that mask. The 0.35-radian threshold in the code defined event duration only. Equation (3) therefore describes the estimator that actually generated the result: finite phase entries were retained without the sketched quality gate. This is a protocol-to-execution discrepancy, not a post hoc reason to revise the old outcome. A quality-gated estimator would be a separately named successor requiring prospective freezing and a new evaluation.

For eligible unit u and fixed projection seed q , let D be mean held-out Poisson deviance per eligible time bin, where lower is better. Define the primary increment

$$\Delta_{uq} = D_{uq,\text{raw}} - D_{uq,\text{PWD}}, \quad (4)$$

so a positive value favors the PWD branch. The three seed-level values are reduced within unit:

$$\Delta_u = \text{median}_{q \in \{1,2,3\}} \Delta_{uq}. \quad (5)$$

Units are then reduced within session j :

$$\Delta_j = \text{median}_{u \in j} \Delta_u, \quad (6)$$

and the development summary is

$$\tilde{\Delta} = \text{median}_{j=1}^{14} \Delta_j. \quad (7)$$

This aggregation makes the session, not the unit or 20-ms time bin, the inferential unit. In the analyzed set, the 14 sessions also correspond to 14 unique specimen identifiers. The statistics remain development summaries because the final set has not been opened.

With all 14 nonzero session effects in the same direction, the two-sided exact sign-test value is

$$p_{\text{sign}} = 2 \binom{14}{0} 2^{-14} = 2^{-13} = 0.0001220703125. \quad (8)$$

Equation (8) describes directional consistency in this selected development set. It is not a confirmatory population claim.

5. Dataset and frozen execution

The source is the Allen Institute Visual Coding Neuropixels dataset described by Siegle and colleagues and distributed as DANDI 000021, immutable version 0.251116.2246 [14,15]. No new animal experiment was performed. The route selected sessions containing exact-acronym LGd and VISp coverage. The LGd-to-VISp ordering was declared from anatomy and the protocol, not learned from the primary outcome; mouse mesoscale tracing supplies independent background for thalamocortical pathway structure [16].

Twenty-one sessions passed the metadata screen. Five were sealed as the final test and 16 entered development. Two development sessions lacked finite receiver coordinates required by Equation (3) and were excluded without coordinate imputation before their primary outcomes were computed. Fourteen sessions from 14 unique specimens remained.

The target was the next 20-ms VISp spike count for each eligible receiver unit. Each session used the first contiguous 80% of eligible time bins for fitting and the final contiguous 20% for evaluation. Standardization and every learned transformation used only the training partition. The analyzed set contained 921 units and 908,295 eligible 20-ms time bins: 726,635 fitting bins and 181,660 evaluation bins.

At each time bin, the raw input contained 14 lag variables: two selected LFP channels at sample offsets 0, 1, 2, 5, 10, 25, and 50 on the 500-Hz resampled grid. Train-only standardization followed by fixed tanh random projections transformed those variables into 16 shared raw-derived features supplied identically to both branches. The capacity-matched raw branch added eight separately seeded projections of the same 14 raw variables. The PWD branch instead added eight projections of the 25-feature encoding in Equation (3). Both models therefore used hidden width 24, 25 trainable Poisson-output coefficients including the intercept, ridge penalty 3.0, and the same three projection seeds.

The defective 12-unit comparator-v1 shakedown and the repaired 12-unit comparator-v2 shakedown are both preserved, but neither entered the grouped estimate. After comparator v2 was frozen, the same predeclared development session, 750749662, was rerun with all 52 eligible VISp units. That full-session result is included once among the 14 session-level aggregate inputs [10].

6. Results

6.1 Primary development gate

All 14 session medians were negative.

Table 2. Primary grouped development result. Deviance values are mean held-out Poisson deviance per eligible time bin; lower is better.

Development summary	Result
Coordinate-eligible sessions	14
Unique specimen identifiers	14
Sessions excluded for undefined coordinates	2
Eligible VISp units	921
Eligible 20-ms time bins	908,295
Fitting / evaluation time bins	726,635 / 181,660
Projection seeds per unit	3
Median raw-minus-PWD deviance	-0.001099
Mean raw-minus-PWD deviance	-0.001114
Session-bootstrap 95% interval	[-0.001301, -0.000872]
Sessions favoring PWD	0/14
Two-sided exact sign-test value	0.000122

The interval is a deterministic 20,000-resample session-level percentile bootstrap with seed 20260719. There were no zero session effects. The negative direction means that the complete PWD branch performed worse than the capacity-matched additional-raw-feature branch in every analyzed session.

6.2 Destruction controls and ablations

Table 3. Within-model destruction controls and component ablations.

Comparison	Median across sessions	95% interval	Direction	What the comparison establishes
PWD minus within-block circular shift	-0.000362	[-0.000398, -0.000278]	14/14 negative	The intact full feature family beat its shifted version. This is a within-model alignment sensitivity, not proof that phase adds value beyond raw input.
PWD minus source-receiver role swap	-0.000158	[-0.000206, -0.000117]	14/14 negative	The original declared assignment beat the role-swapped version. This does not establish anatomical or causal direction.
PWD minus phase ablation	+0.000189	[+0.000136, +0.000210]	14/14 positive	Removing the configured phase terms improved prediction.
PWD minus amplitude-duration ablation	-0.000154	[-0.000219, -0.000123]	14/14 negative	Removing amplitude-duration terms worsened the already losing PWD branch. This does not prove standalone incremental value over raw features.

Negative PWD-minus-null deviance favors intact PWD. Each all-one-direction comparison has the same descriptive exact sign-test value of 0.000122. The four control sign tests are descriptive and were not adjusted for multiplicity. These controls expose structure inside a failed model; they do not overturn the primary gate. The most defensible reading is:

- temporal arrangement mattered relative to the specified shifted null;
- the declared source-receiver assignment mattered relative to a role swap;
- amplitude-duration terms contributed inside the configured branch;
- the configured phase terms degraded held-out prediction;
- the complete PWD branch still lost to the equally sized raw branch.

This result is narrower than Montemurro et al. [1]. It uses a different species, preparation, stimulus set, pathway, target, phase construction, and comparator. It therefore rejects neither phase-of-firing coding in general nor every receiver-relative temporal operator. It rejects promotion of this frozen 25-feature implementation for this bounded prediction task.

7. NAPOT decision vector

The component result should be recorded as a vector of decisions rather than a binary verdict.

Table 4. NAPOT component-gate decision vector.

Gate	Question	Current status
G0	Is the open dataset, version, split, and preprocessing route reproducibly identified?	Passed for this bounded development analysis.
G1	Are the tested features computed without future-sample access?	Software-verified causal computation; not an empirical causality result.
G2	Does the full PWD branch improve held-out neural prediction over matched raw capacity?	Failed in 14/14 development sessions.
G3a	Does the intact feature family beat its temporal-shift null?	Within-model supported in 14/14 sessions; not phase-specific incremental support.
G3b	Does the original declared edge beat source-receiver role swap?	Within-model supported in 14/14 sessions; not proof of anatomical or causal direction.
G4	Do the configured phase components add value?	Failed; phase ablation improved prediction in 14/14 sessions.
G5	Does a full recurrent NAPOT model beat the frozen M0-M6 alternatives?	Not tested.
G6	Does the model predict action and refference, meaning returned sensory consequences of its own actions?	Not tested.
G7	Does phase-specific intervention change receiver transfer as predicted?	Not tested.
G8	Does any decoded state correspond to perceptual content or consciousness?	Not tested.

These data support blocking the tested component from promotion. They do not license total rejection of the larger conjunction. The surviving within-model controls also do not validate PWD or NAPOT because the complete branch lost its primary matched comparison.

8. Consequences for the research program

The result changes the next NAPOT benchmark in five enforceable ways.

1. **The adverse implementation remains permanent.** Every successor must cite the 14-session result and preserve its exact numerical and procedural bindings.
2. **A redesigned phase estimator is a new estimator.** Implementing the parent protocol’s previously omitted phase-quality mask, using stricter phase-quality rules, independently localizing delays, adding waveform-shape controls, or separating local phase departure from spatial propagation may be tested only in a separately named and prospectively frozen protocol. None may be retrofitted into the failed implementation.
3. **Amplitude-duration structure needs its own fair branch.** Its apparent within-model contribution must be compared directly with an equal-input raw temporal model rather than inferred from an ablation inside a losing branch.

4. **Direction needs independent evidence.** The anatomical declaration [16] and role-swap sensitivity answer different questions. A causal direction claim requires an independently localized lag or an intervention that changes receiver transfer as predicted.
5. **The five held-out sessions stay sealed.** They may be used once for the unchanged failed development model only after a recorded methods decision, or reserved for a separately named successor. Development-driven redesign and unchanged-model confirmation cannot share the same final claim.

The full NAPOT benchmark remains a separate prospective experiment. It requires spikes, LFP, stimulus and trial tables, movement, eye variables, spatial metadata, action outcomes, and subject-level isolation. Its complete M0-M7 ladder must match latent dimension, capacity, optimization budget, splits, and seeds [9]. Neural prediction must be established before behavior is used to promote the architecture.

9. Nonduplication and publication role

Phase-Wave Differentials: A Receiver-Relative Signal Operator for Testing Neural State Updates is the empirical source. This paper must never be counted as:

- a second analysis of an independent cohort;
- a replication;
- a second positive or negative empirical result;
- evidence that the final holdout was opened;
- evidence for the full NAPOT architecture.

This paper is publishably distinct only as a brief report or program update that supplies a formal inference boundary and an auditable decision vector. Its contribution is governance: it states which claims must be withdrawn, which remain open, which within-model observations may guide redesign, and which data must remain sealed. If released, this paper should cross-reference *Phase-Wave Differentials: A Receiver-Relative Signal Operator for Testing Neural State Updates* as the sole execution and should not duplicate the companion empirical paper’s complete methods presentation.

10. Limitations

The preparation is passive visual coding in mice. The endpoint is short-horizon VISp spike count, not image identity, behavior, report, action, reafference, subjective content, or consciousness. The feature set contains one declared LGd-to-VISp channel pair per session and one fixed operator family. It does not instantiate the complete multisite, multiscale, dynamically gated recurrent architecture described by NAPOT.

The random-feature Poisson comparator is deliberately capacity matched but is not the strongest conceivable neural sequence model. The within-session development split is contiguous, whereas a future confirmatory benchmark with repeated specimens should isolate animals across folds. Fourteen unique specimen identifiers strengthen the independence of the present session summaries, but the development set remains selected for model construction and cannot substitute for the sealed final set.

The phase branch may be distorted by nonsinusoidal waveform shape and nonstationarity, while phase-connectivity estimates are additionally vulnerable to volume conduction or source mixing, noise, and sample-size bias [2-7]. The executed estimator did not apply the amplitude / cycle / concentration quality mask sketched by its parent protocol. Low-amplitude phase reliability, reference and filter sensitivity, and lag misspecification are therefore method-specific contingencies requiring explicit controls in any successor. The coordinate-normalized phase-gradient term is not a physical propagation velocity.

The bootstrap and sign test describe consistency across development sessions. They do not license confirmatory population inference. No result here establishes that a neural manifold is a viewed internal picture, that a PWD is a unit of qualia, or that a recurrent reconstruction is conscious.

11. Claim-source-genealogy atlas

The accompanying paper-local atlas binds each substantive claim to:

- the exact manuscript locator;
- its claim class and evidence boundary;
- the empirical aggregate or dated Micah source;
- its earliest bounded ancestor and first mature formulation;
- named earlier scientific background;
- a failure condition;
- provisional corpus atom identifiers for later unified fan-in.

The atlas keeps genealogy and scientific evidence separate. A public date proves that wording existed by that date; it does not prove a neural mechanism. An empirical adverse result narrows the tested mechanism; it does not erase the dated genealogy. The global atom identifiers remain provisional until the 77-paper corpus-wide deduplication pass.

12. Conclusion

The decisive result is simple: the frozen 25-feature receiver-relative branch did not improve held-out prediction. The capacity-matched raw branch won in every one of 14 development sessions. Removing the configured phase terms also improved prediction in every session.

The controls make the failure useful rather than ambiguous. Temporal arrangement, the declared source-receiver assignment, and amplitude-duration terms affected performance inside the proposed branch, but none of those within-model findings overcomes the failed primary comparison. The result blocks this unmasked implementation from being cited as empirical support for NAPOT while preserving a concrete redesign program.

The broader question remains open: can distributed, directionally constrained, temporally structured projections improve recurrent neural-state reconstruction, predict action and refference, and survive causal intervention? A credible answer must begin with this adverse result, preserve it, and test a prospectively frozen successor rather than rewriting the failed model after seeing the outcome.

Authorial provenance and research chronology

Artificial intelligence was not involved in originating the core SAN, PWD, or NAPOT ideas. Micah Blumberg developed the underlying research program across sources spanning 2011-2026, including public audio in 2017 and versioned public Git records in 2022. Later AI tools assisted in bounded roles including mathematical formalization, code, editing, source recovery, and literature checking. Micah Blumberg selected the claims and sources, directed the analysis, and remains responsible for the arguments, interpretations, and errors.

Data, code, and custody availability

No new model was fitted and no new dataset was analyzed for this paper. The analyzed data are DANDI 000021, immutable version 0.251116.2246 [15]. The companion empirical paper owns the execution code and results. This paper binds its numerical statements directly to the grouped aggregate with SHA-256 C5C48C9802DBA4349601ABF0D44EFFDE1261666DD8C516D17022B223B2BC63F7 and its validation receipt with SHA-256 09DFC47DAA71EBA678EF4D4D2A88EA0F2E6BAFA00DC519B80E0BE14A08BFED6. The executed feature path is independently bound to `causal_signal_operators.py` SHA-256 510D23F11B3E5D168156324A486D790B7B0C8B3440F4C04A410A05F12340272E and `run_development_session_v2.py` SHA-256 1E06781D3DC2009E246F430D7783B979EA60C9E315385B671D038044E4F1D19F. This avoids treating later edits to companion prose as changes to the frozen empirical record. Five final-test sessions remain unopened.

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